

Jiadong Lu

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EDUCATION

Zhejiang University

Master's degree in Control Science and Engineering

- FASTLab. Advised by [Chao Xu](#) and [Yanjun Cao](#)

China

Sep. 2024 – Present

Shandong University

Bachelor's degree in Control Science and Engineering

- GPA: 4.0/4.0 (Rank: 2/28)

China

Sep. 2020 – Jun. 2024

PUBLICATIONS & PREPRINTS

[1] **Parallel Continuous-Time Relative Localization with Augmented Clamped Non-Uniform B-Splines** [\[Paper\]](#)

Jiadong Lu, Zhehan Li, Tao Han, Miao Xu, Chao Xu, Yanjun Cao. *Submitted to T-RO, under second-round review.*

[2] **Hierarchical Cooperative Relative Localization via Closed-Form Solutions and Relative Kinematics** [\[Paper\]](#)

Z. Li, J. Lu, Z. Wang, Z. Xun, C. Xu, F. Gao, Y. Cao. *T-RO, conditionally accepted.*

[3] **NeuRIO: A Streaming Neural Estimator for Zero-Shot Sim-to-Real Multi-Robot Relative Inertial Odometry**

Zhehan Li, Jiadong Lu, Shengwei Ren, Chao Xu, Yanjun Cao. *Submitted to ICRA 2027.*

[\[Paper\]](#)

[4] **Mr. Virgil: Learning Multi-robot Visual-range Relative Localization** [\[Paper\]](#)

S. Wang, Z. Li, J. Lu, R. Xiong, Y. Cao, Y. Wang. *IROS 2025, Oral.*

[5] **Novel Design of Reconfigurable Tracked Robot with Geometry-Changing Tracks** [\[Paper\]](#)

C. Xuan, J. Lu, Z. Tian, J. Li, M. Zhang, H. Xie, J. Qiu, C. Xu, Y. Cao. *IROS 2024, Oral.*

ONGOING WORKS

[W1] **FACT-LIVO: Fast Continuous-Time Lidar-Inertial-Visual Odometry and Mapping** [\[Demo\]](#)

Jiadong Lu, Zhehan Li, Jiajun Lv, Chao Xu, Yanjun Cao. *In preparation.*

SELECTED RESEARCH EXPERIENCE

NeuRIO: Streaming Neural Multi-Robot Relative Inertial Odometry

Nov. 2025 – Sep. 2026

- Developed a streaming multi-robot estimator that combines node-factor cross-attention and GRU memory to infer 6-DoF relative poses from bearings, ranges, and IMU measurements without iterative optimization.
- Training with simulated sensor streams and temporally persistent corruptions enabled zero-shot transfer, yielding mean sequence-level RMSEs of 14.1 cm in position and 3.9° in rotation across 24 real-world sequences.
- The model supports different team sizes and time-varying observation graphs, with inference below 20 ms per update for up to 400 simulated robots on an RTX 4070 Ti SUPER GPU.
- Paper [3] submitted to ICRA-2027.

Continuous-Time LiDAR-Inertial-Visual Odometry and Mapping

Feb. 2026 – Present

- Developed a continuous-time SLAM system based on Clamped Non-Uniform B-splines (C-NUBS).
- Proposed closed-form prior transport to resolve Schur-complement prior inconsistency induced by C-NUBS extension, enabling consistent long-term estimation.
- Derived dense continuous-time PGO that preserves full trajectory information with near-discrete-PGO efficiency.
- Developed a general framework that accelerates continuous-time spline optimization through shared-support factor batching and direct normal-equation assembly.
- Manuscript [W1] in preparation, targeting IEEE Transactions on Robotics.

SELECTED RESEARCH EXPERIENCE (continued)

Continuous-Time Multi-Robot Direct Relative Localization

Nov. 2024 – Feb. 2026

- Introduced a C-NUBS-based continuous-time state representation, eliminating the query-time latency of conventional B-spline formulations.
- Formulated a continuous-time factor graph for relative localization, enabling natural fusion of asynchronous multi-robot observations and online inter-robot time-offset estimation.
- Developed a robot-wise asynchronous block coordinate descent solver, enabling real-time optimization of multi-robot relative states.
- Paper [1] submitted to IEEE Transactions on Robotics (under second-round review).

Hierarchical Multi-Robot Direct Relative Localization

May. 2024 – Nov. 2025

- Designed an online, robust, and fast hierarchical initialization method, which can overcome over 90% outliers and obtain accurate states of all neighbors in a time window.
- Developed a real-time tightly-coupled sliding-window optimizer that fuses multi-robot IMU preintegration, UWB distances, and camera bearings under robocentric relative kinematics for NLOS scenarios.
- Paper [2] has been conditionally accepted by IEEE Transactions on Robotics.

Learning Multi-robot Visual-range Relative Localization

Nov. 2024 – Mar. 2025

- Co-developed an end-to-end multi-robot UWB-visual relative localization framework.
- Designed a multiplex graph attention network with self- and cross-edge message passing, coupled with differentiable Sinkhorn assignment and a PGO back-end for joint front-end-back-end training.
- Paper [4] accepted by IROS-2025 as an oral presentation.

PROJECT EXPERIENCE

Air-Ground Cooperation without Global Information [\[Video\]](#)

Zhejiang, China

IROS 2025 EXPO | Team Leader

2025

- Live demo showcasing autonomous collaboration between RoFly, an amphibious quadrotor, and CubeTrack [5], a reconfigurable tracked vehicle, enabled by our relative localization systems [1, 2, 4] and a non-inertial MPC controller without global information.

HONOURS AND AWARDS

Presidential Scholarship Finalist, Shandong University	2023	Top 1%
Outstanding Graduate Student, Zhejiang University	2026	Top 10%
Excellent Graduation Thesis, Shandong University	2024	Top 5%
First Prize, National Intelligent Car Competition, Chinese Association of Automation	2022	Top 5%
First Prize Scholarship, Shandong University	2022 & 2023	Top 5%

SKILLS

Programming: C++ (6+ years), Python (6+ years), MATLAB (5+ years).

Frameworks & Libraries: ROS (4+ years), Eigen, Ceres, PyTorch, OpenCV, Open3D.

Development & Deployment: Linux, IsaacSim, Git, Docker, SolidWorks, Fusion, LaTeX.